SWIGGY SALES ANALYSIS

1. BUSINESS GOAL:

To use data analytics to optimize restaurant performance, improve customer satisfaction, and support strategic decisions in a food delivery ecosystem.

2. DATA SUMMARY:

Kaggle – Swiggy Bangalore Restaurant Dataset

Rows: - 8,600 restaurants

Columns: - State, Restaurant Name, Area, City, Average Price for Two, Average Rating, Total Ratings (Votes), Food Types (Cuisines), Delivery Time (Minutes).

3. EXPLORATORY DATA ANALYSIS USING PYTHON:

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.
- Missing Data Handling
- Column Standardization
- Formatted Numerical values of Price and Avg_Ratings Columns
- Created new columns as follows: -
  - Price_Category Column based on dataset of Price Columns
  - Delivery_Status Column based on dataset of Delivery_Time Columns
  - Rating_Status Column based on dataset of Avg_Ratings Columns
  - Popularity_Score Column based on Avg_Rating and Total_Rating
  - Popularity_Status Column based on dataset of Popularity_Score Columns

4. DATA ANALYSIS USING SQL:

1. <u>Restaurant Count By Area: -</u> Counted number of restaurant in each area.

Data Output Messages Notifications

Showing rows: 1 to 843

	City text	Area text	Restaurant_Count bigint
1	Delhi	Rohini	257
2	Mumbai	Chembur	208
3	Pune	Kothrud	149
4	Mumbai	Andheri East	135
5	Ahmedabad	Navrangpura	132

Total rows: 843 Query complete 00:00:01.589

2. Restaurant with bad popularity score: - Counted number of restaurants having bad popularity score in each area.

Data Output | Messages | Notifications

SQL

	City text	Restaurant_Count bigint
1	Mumbai	697
2	Pune	640
3	Kolkata	632
4	Chennai	469
5	Hyderab...	446
6	Ahmeda...	403
7	Bangalore	403

Total rows: 9 | Query complete 00:00:00.319

3. Restaurant Count with Bad Delivery: - Counted number of restaurants in area having bad delivery time.

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 386

	City text	Area text	Restaurant_Count bigint
1	Ahmedab...	Vastral	10
2	Ahmedab...	Kuber Nagar	7
3	Ahmedab...	Chandkheda	7
4	Ahmedab...	Isanpur	4
5	Ahmedab...	Narolgam	2
6	Ahmedab...	Memnagar	2
7	Ahmedab...	Gidc Naroda	2

Total rows: 386 | Query complete 00:00:00.886

4. Best Restaurant Count: - Counted Number of Restaurants with high rating and popularity score.

Data Output | Messages | Notifications

SQL | Showing rows: 1 to 199

	City text	Area text	Restaurant_Count bigint
1	Ahmedab...	Navrangpura	4
2	Ahmedab...	Ellisbridge	3
3	Ahmedab...	Vastrapur	3
4	Ahmedab...	Ahmedabad	1
5	Ahmedab...	Maninagar	1
6	Ahmedab...	Naranpura	1
7	Bangalore	Ashok Nagar	11

Successfully

Total rows: 199 | Query complete 00:00:00.168

5. <u>Restaurant At Risk: -</u> Counted number of restaurants with high average rating and less number of rating.

Data Output | Messages | Notifications

	City (text)	Restaurant_Count (bigint)
1	Kolkata	218
2	Chennai	214
3	Mumbai	209
4	Pune	167
5	Bangalore	162
6	Ahmedab...	125
7	Hyderabad	124

Total rows: 9 | Query complete 00:00:00.148

6. <u>Top Food by City and Area: -</u> Checked the best selling food in each city and area and counted the number of cuisine.

Data Output | Messages | Notifications | Showing rows: 1 to 10

	City (text)	Area (text)	Primary_Cuisine (text)	Cuisine_Count (bigint)
1	Ahmedabad	Akhbar Nagar Circ...	Gujarati	1
2	Ahmedabad	Acher	North Indian	1
3	Ahmedabad	Ahmedabad	Indian	6
4	Ahmedabad	Ambavadi	Chinese	1
5	Ahmedabad	Ambawadi	North Indian	5
6	Ahmedabad	Bapunagar	North Indian	1
7	Ahmedabad	Bhadra	Indian	1

Total rows: 10 | Query complete 00:00:00.167

7. <u>Restaurant Count with all rating type: -</u> Counted the number of restaurants with all the three ratings in each city.

Data Output | Messages | Notifications | Showing rows: 1 to 27

	City (text)	Rating_Status (text)	Restaurant_Count (bigint)
1	Ahmedab...	Low Rated	304
2	Ahmedab...	High Rated	263
3	Ahmedab...	Avg Rated	150
4	Bangalore	High Rated	474
5	Bangalore	Low Rated	311
6	Bangalore	Avg Rated	161
7	Chennai	High Rated	552

Total rows: 27 | Query complete 00:00:00.163

8. <u>Top 10 Popular Restaurant: -</u> Looked for the top 10 most popular restaurants as per popularity score.

Data Output Messages Notifications

Showing rows: 1 to 10

	City text	Area text	Restaurant text	Popularity_Score double precision
1	Mumbai	Sion	Guru Kripa (Sion West)	38.328
2	Hyderabad	Koti	Grand Hotel	37.763
3	Hyderabad	Kavadiguda	4M Biryani House	37.763
4	Hyderabad	Chintal	Lucky Restaurant	36.842
5	Hyderabad	Toli Chowki	Shah Ghouse Cafe & Restaura...	36.842
6	Delhi	Pitam Pura	Gulab Wala	36.625
7	Kolkata	Kalighat	Bachan'S Dhaba	36.625

Total rows: 10 Query complete 00:00:00.167

9. <u>Area with best rating: -</u> Counted the number of restaurant in area having best average rating.

Data Output Messages Notifications

Showing rows: 1 to 227

	City text	Area text	Avg_Ratings double precision	Restaurant_Count bigint
1	Chennai	Alwarpet	4.6	7
2	Bangalore	Ashok Nagar	4.7	6
3	Chennai	Anna Nagar	4.6	5
4	Pune	Koregaon Park	4.6	5
5	Mumbai	Santacruz East	4.6	4
6	Delhi	Rohini	4.6	4
7	Mumbai	Santacruz East	4.7	4

Total rows: 227 Query complete 00:00:00.199

10. <u>Restaurant with high price and low popularity score: -</u> Looked for the restaurants who are at risk based on the price and popularity score.

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1

	City text	Restaurant text	Price_Category text	Popularity_Score double precision
1	Mumbai	The Spot.	Costly	8.525
2	Ahmedabad	Tandoorworkz	Costly	8.525
3	Hyderabad	Zouq Fusion Foods	Costly	8.525
4	Mumbai	Mughal Darbar	Costly	8.525
5	Chennai	Coal Barbecues Express	Costly	8.525
6	Delhi	Pizzallo Hot	Costly	9.134
7	Kolkata	New Maa Kali South Indian Food Centre		

Successfully run. Total query runtime: 171 msec. 26

Total rows: 2625 Query complete 00:00:00.171

11. Cuisines tend to be premium: - Selected the cuisines of each city that are premium based on the popularity score and price category.

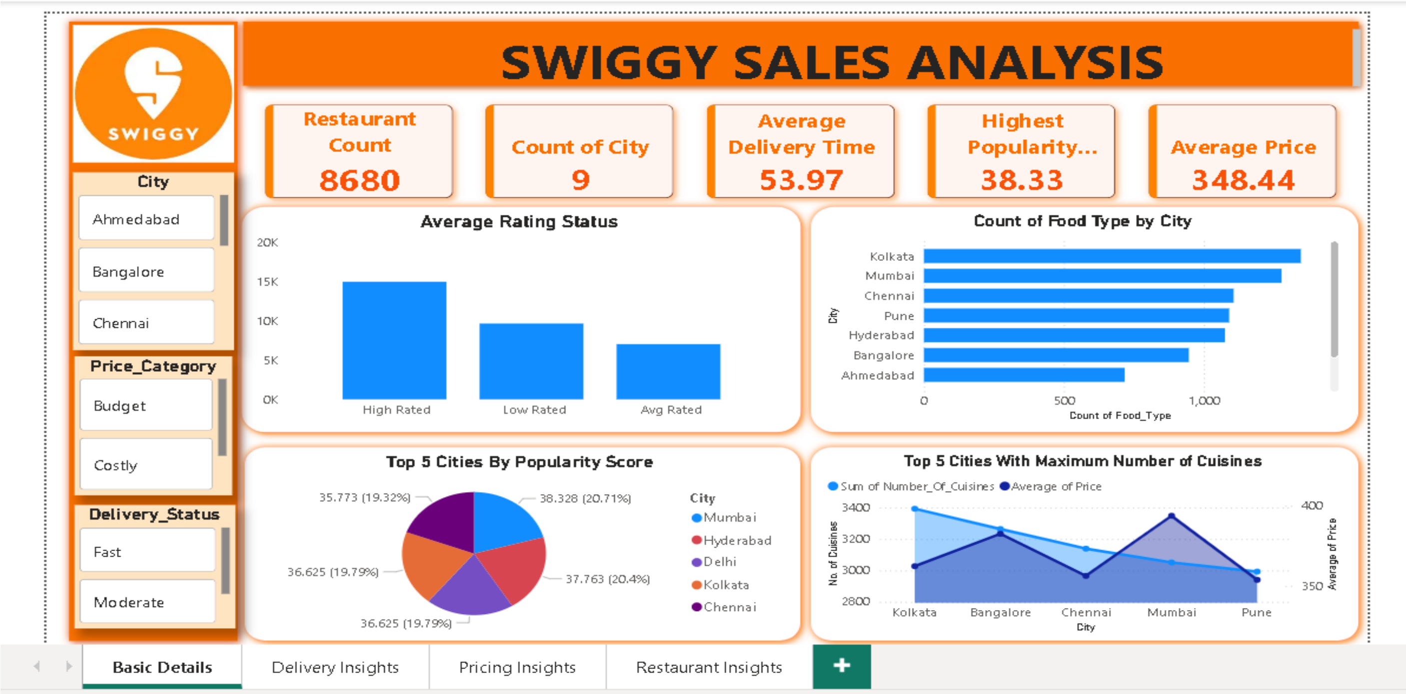
Data Output | Messages | Notifications

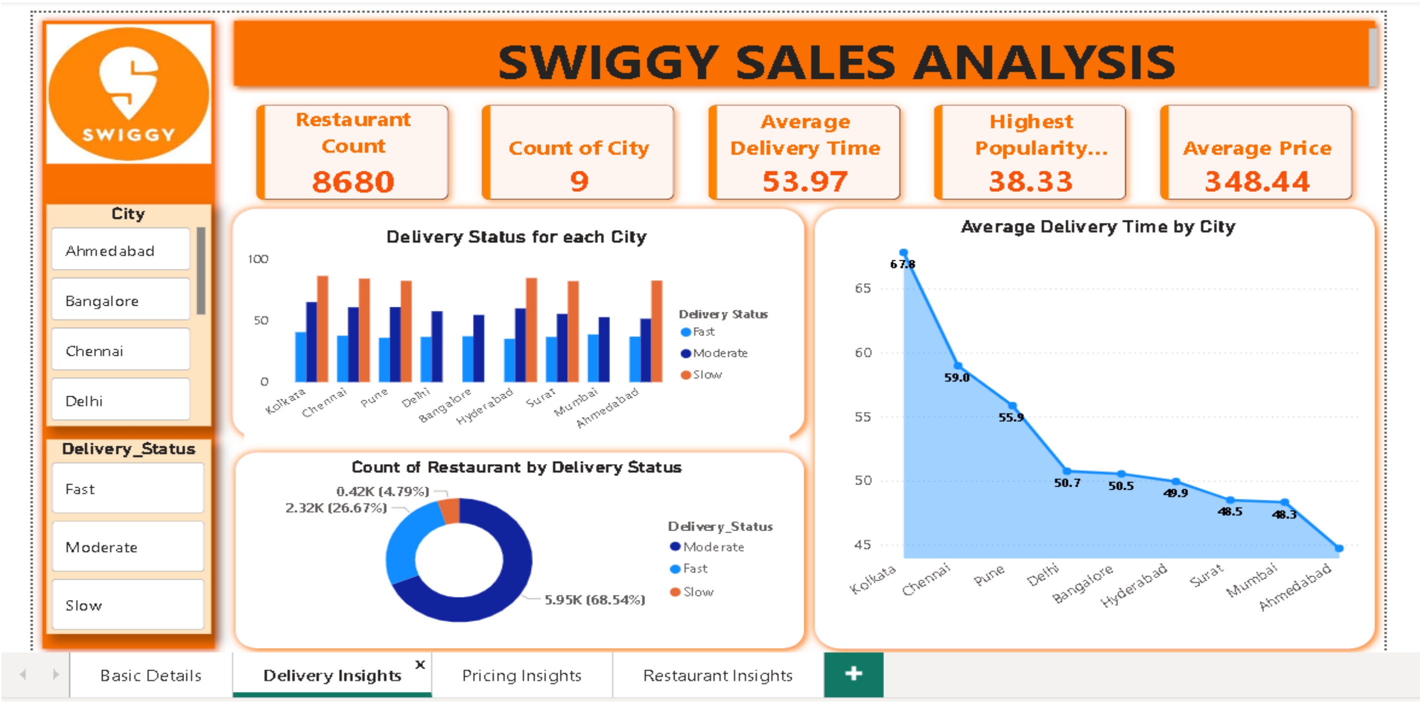
Showing rows: 1 to 10 | Page No: 1

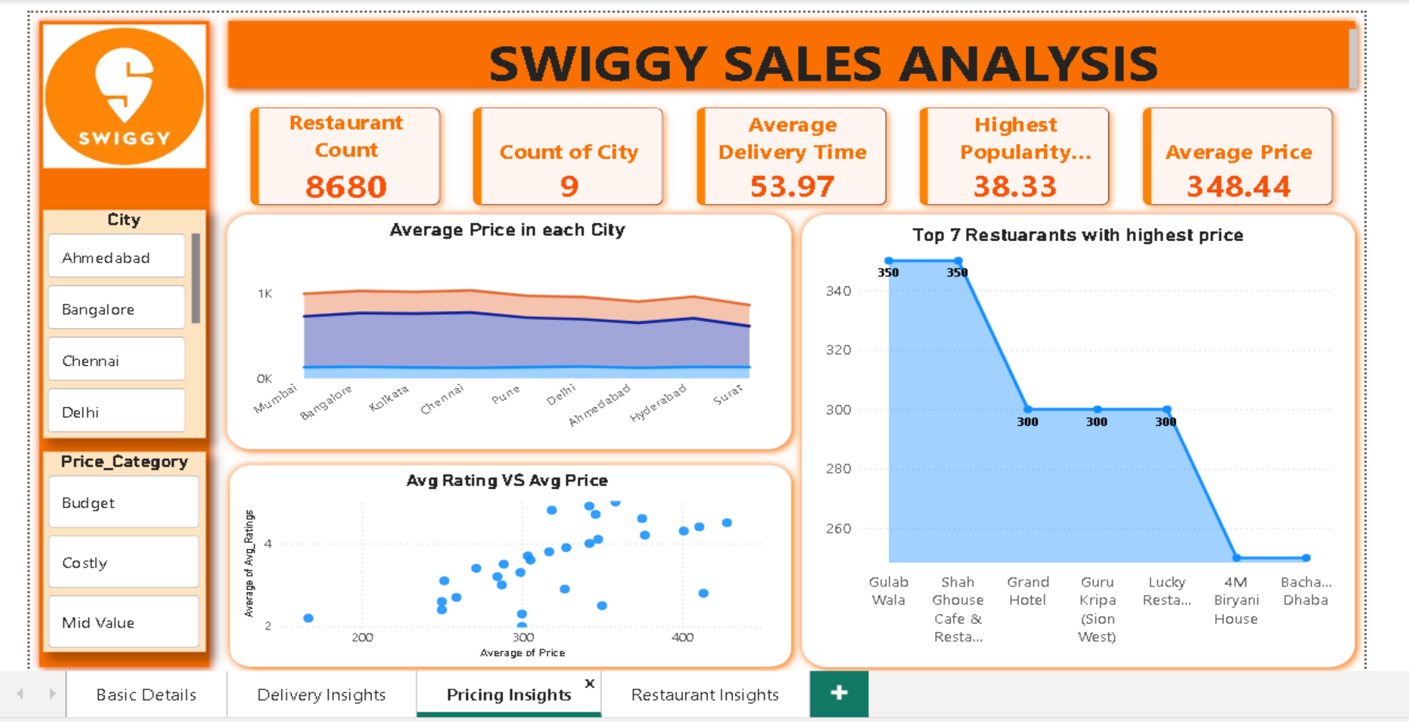
	Primary_Cuisine (text)	Avg_Price (double precision)	Restaurant_Count (bigint)
1	Chinese Pan-Asian Tibetan Oriental Mongolian	1800	1
2	Japanese	1300	17
3	Tribal	1200	1
4	Indian Seafood Chinese Ice Cream	1200	1
5	Goan	1150	2
6	Italian American Mexican Pastas Pizzas Desserts Salads Snacks	1100	1
7	Asian European Italian	1100	1

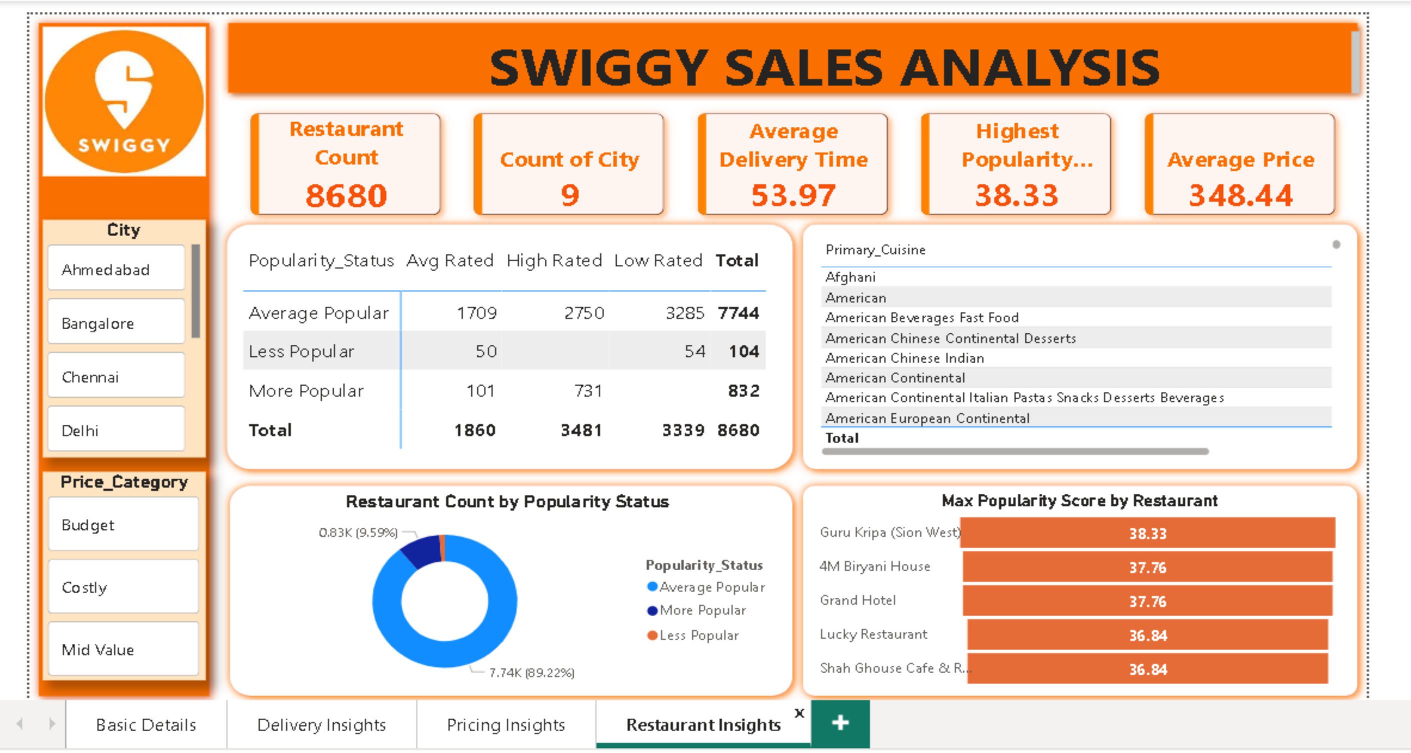
Total rows: 10 Query complete 00:00:00.131

5. POWER BI DASHBOARD:









6. BUSINESS RECOMMEDATIONS:

- **Fix Slow Deliveries**: Add more delivery riders in high-delay areas like Ahmedabad, Kolkata, and Chennai to cut down on long wait times.
- **Boost "Hidden Gems":** Promote "At Risk" restaurants (those with high ratings but very few reviews) on the app's homepage to help them get more orders.
- **Help Expensive, Low-Popularity Places:** Offer special discounts or meal combos for the 2,625 costly restaurants that are struggling to get customers.
- **Expand Premium Food:** Encourage new restaurants to sell high-profit, premium foods like Japanese or Pan-Asian cuisine in highly-rated areas like Ashok Nagar (Bangalore).
- **Improve Low-Rated Spots:** Provide training and quality checks to the large number of low-rated budget restaurants to stop customers from leaving the platform.